European Commission

Defence Industry and Space

Multi-Domain Combat Clouds

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DEFENCE APPLICATIONS

Improved data collection and communication capabilities of defence technologies makes integration across domains possible

Multi-Domain Combat Clouds can dramatically improve situational awareness at all levels of command

The increase in data transmitted between endpoints makes cybersecurity a priority

Why do MDCCs matter for defence?

The Multi-Domain Combat Cloud (MDCC) is a secure network of connected battlefield assets that integrates data from the air, sea, land, and space domains into a **comprehensive picture of the battlefield**, thus improving collaboration between actors on the strategic, operational, and tactical level.

As battlefield technology becomes increasingly capable of collecting and transmitting data, the potential advantages to be gained from **leveraging data for insights** can translate into strategic as well as tactical advantages.

In the short term, better situational awareness enabled by the **integration of data sources** from the battlefield translates to better decision-making. On a longer timescale, the collection of various types of data can be used to inform military doctrine and as **input for training military AI systems** on proprietary data. Investing into developing an interoperable set of technologies now can have large returns in terms of industry capability to deliver an increasingly integrated network of defence technologies.

A key use-case of MDCCs is real-time coordination between naval, air, ground and space assets, which improve effectiveness in areas such as targeting and intelligence gathering.

What does the EU aim to achieve?

Since the launch of the European Defence Fund (EDF), the development of Multi-Domain Combat Cloud technologies has been a strategic priority. **Information Superiority** is one of the key categories of action, with more than 10% of the EDF budget expected to be allocated to this area.

The EDF Indicative Multiannual Perspective 2025–2027 prioritises the development of military operational cloud systems operating at multiple scales and security levels. The effective exploitation of data generated by these systems will also underpin the **development of European defence artificial intelligence (AI) solutions**, trained and optimised using data collected through MDCC environments.

Main advantages of using MDCCs



Increased situational awareness



Improved collaboration



Training data generation for AI applications



Real-time plan adaptions



Better post-mission insights



Higher planning precision

Main vulnerabilities arising from the use MDCCs



Overreliance on proprietary systems



Cybersecurity vulnerabilities



More complex training demands



Costly retrofitting



Data security issues



Inconsistent interoperability

When will EDF project results be translated into military capabilities?

Through the European Defence Fund, the EU has financed multiple projects supporting the goal of developing MDCCs.

A flagship example is the EDOCC (European Defence Operational Collaborative Cloud) project, selected under the EDF 2021 call. EDOCC focuses on the study and design of a **virtual platform enabling the integration of military technologies** while identifying the standards necessary for future MDCC development.

Other examples of key EDF projects include **EDINAF**, which is developing a naval operational cloud; **SEQURED**, focused on quantum-resistant data sharing; **PROTEAS**, which designs a multi-environment command post for special operations; **OPTIMAS**, exploring laser communication system integration; and **SESIOP**, advancing interoperability in command-and-control architectures.

Through such projects, the EU is reinforcing the technological edge and operational readiness of the European defence industry and armed forces in an increasingly digitalised and network-centric operational environment.

EDF projects delivering initial results



SESIOP
Improve the interoperability of air command and control
EU financing: €18.8M



EDINAF
Joint naval operational cloud
EU financing: €29M



EDOCC
European Defence Operational Collaborative Cloud
EU financing: €40M

EDF projects expected to deliver results in 2026-2028



OPTIMAS
Airborne laser communication system integration
EU financing: €16.9M



PROTEAS
Special operations forces multi-environment command post
EU financing: €20M



SEQURED
Quantum-resistant data sharing
EU financing: €4M

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